# 10 • Quality Assurance — Risk-Based Testing

> **Assignment:** *“Quality Assurance Testing: Design an approach to risk-based testing. Conduct tests to identify L2 – L4 translation issues. Address any bugs or issues identified during testing."*

> **Lecture Day 5:** *“We cannot test everything, so we test what matters most."*
>
> 1. Identify Hazards → 2. Rank by Risk → 3. Test Critical Paths First → 4. **Document Residual Risk**

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## 10.1 Testing Approach

We follow the four-step risk-based procedure from Day 5. The depth of testing is determined by the **potential for clinical harm**, not by the number of rules.

### Types of Testing Used

| Type (Day 5) | In our project |
|---|---|
| **L2 Verification** | Reconciliation of decision table ↔ BPMN gateway cascade ↔ L3 rules: do the conditions, the ordering and the outputs match? |
| **Functional Testing** | Individual rules, exclusions, severity ordering, boundary values |
| **Integration Testing** | Interplay: derived elements → rules → output; conditional logic → rule reachability; trend rules across several days |
| **Usability Testing** | Image assignment by non-clinical persons — **not yet carried out** (see residual risk) |

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## 10.2 Steps 1 & 2 — Hazard Analysis and Prioritisation

| ID | Hazard | Origin | Severity (1–5) | Likelih. (1–5) | Score | Mitigation | Covered by |
|---|---|---|---|---|---|---|---|
| **H-01** | An infection is classified as GREEN | The rule does not fire; the question is not asked because of conditional logic | 5 | 3 | **15** | Worst-first cascade; safety-critical questions ungated; catch-all only as the final rule | T1–T5, T11–T13 |
| **H-02** | An emergency is classified as ORANGE instead of RED | Prioritisation incorrect; rule order transposed | 5 | 2 | **10** | Fixed worst-first ordering, documented and tested | T4, T15 |
| **H-03** | The patient abandons the check, leaving a gap in the trend | Too many questions; excessive duration | 3 | 3 | **9** | Baseline data collected only once; conditional logic reduces the number of questions; ≤ 3 min | T6 |
| **H-04** | False alarm during normal healing | Threshold set too low; time window not taken into account | 2 | 4 | **8** | Time-dependent rule R6a; information rules actively reassure | T5, T9, T14 |

| ID | Hazard | Origin | Severity (1–5) | Likelih. (1–5) | Score | Mitigation | Covered by |
|---|---|---|---|---|---|---|---|
| H-05 | An implausible input is processed | Constraint missing or applied too late | 3 | 2 | **6** | Constraints run before rule evaluation (validation loop in the BPMN) | T7 |
| H-06 | The practice overlooks an alert | Alert fatigue caused by too many notifications | 4 | 2 | **8** | Alerts only for ORANGE/RED; structured content instead of free text | *(organisational, not testable)* |

### Risk Matrix

```
Severity
  5 │                   H-02(10)   H-01(15)
  4 │          H-06(8)
  3 │                   H-05(6)    H-03(9)
  2 │                              H-04(8)
  1 │
    └────────────────────────────────────────
      1        2         3          4      5    Likelihood
```

### Step 3 — Prioritisation

- **H-01 and H-02** (score ≥ 10): **full coverage**. Every rule at the red and orange level has at least one test case, and all boundary values are checked from both sides.
- **H-03 to H-06** (score 6–9): **sample-based coverage** with 1–2 representative test cases each.
- The green catch-all rule is only tested indirectly — through cases that are *not* supposed to escalate.

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## 10.3 Defects Found

> All findings originate from **desk-based testing** and a systematic review of the decision table against the L3. **Three of the five defects arose in the L2 and only became apparent during L4 testing — exactly the "L2–L4 translation issues" that the assignment asks about.**

### B-01 · Marked signs of inflammation on days 1–2 are classified as GREEN

| | |
|---|---|
| **Severity** | 🔴 high |
| **Originating layer** | **L2** — decision table |
| **Description** | R6 was gated with `days_post_op >= 3`. A patient on day 2 with a markedly reddened, swollen, overheated wound received **GREEN** via the catch-all rule R8 — provided there was neither fever nor pus |
| **Root Cause** | The time-based exception for *mild* post-operative erythema was inadvertently applied to **all** signs of inflammation. Related to challenge C-03 |
| **Aggravating factor** | Test case T5 originally recorded this behaviour as *expected* — the test would never have found the defect |
| **Fix** | R6 split: **R6a** (mild signs, Southampton I) remains time-dependent · **R6b** (marked signs, Southampton II or `swelling = 3`) fires **from day 1**. T5 corrected |

| | |
|---|---|
| **Retest** | T13, T14 |

### B-02 • “I feel really unwell" triggers nothing

| | |
|---|---|
| **Severity** | 🔴 high |
| **Originating layer** | **L2** — decision table |
| **Description** | `wellbeing` was collected ( `0=Gut · 1=Müde · 2=Richtig krank` — 0=well · 1=tired · 2=really unwell) but appeared in **not a single rule**. A patient with `wellbeing = 2` , 37.6 °C and an unremarkable wound received **GREEN** |
| **Root Cause** | The data element was defined but never carried over into the logic. Clinically relevant: a severe subjective sense of illness can **precede** measurable fever — in early sepsis this is precisely the dangerous pattern |
| **Fix** | New rule **R9**: `wellbeing = 2 OR nausea_vomiting = 2` → ORANGE |
| **Retest** | T11 |

### B-03 • Emergency rule R2 was unreachable because of conditional logic

| | |
|---|---|
| **Severity** | 🔴 high |
| **Originating layer** | **L2/L3** — conditional logic |
| **Description** | The question about the extent of erythema was gated with `relevant: wound_state >= 2` . R2 (spreading erythema / red streak → RED, suspected lymphangitis) could only fire if the patient rated the wound appearance as at least “slightly reddened" ("leicht gerötet"). If they selected “Calm & dry" (“Ruhig & trocken") — plausible when the wound itself is unremarkable but a red streak runs towards the trunk — **the question was never asked and the emergency rule was never reached** |
| **Root Cause** | The conditional logic was set from a UX perspective (fewer questions) without checking the reachability of safety-critical rules. **A textbook case of an L2→L4 translation issue** |
| **Fix** | New, **ungated** data element `red_streak` (DE-032): *“Does a red streak run away from the wound?"* (*“Zieht ein roter Streifen von der Wunde weg?"*). R2 changed to `red_streak = 1 OR redness_extent = 4` |
| **Retest** | T12 |
| **Rule derived from this** | Safety-critical questions must **never** sit behind conditional logic. Adopted as a convention in the L3 |

### B-04 • Duplicated source of truth for the fever threshold

| | |
|---|---|
| **Severity** | 🟡 medium (maintainability) |
| **Originating layer** | **L3** |
| **Description** | The derived element `fever = temp >= 38` existed but was used nowhere; R5 checked `temp >= 38` again directly. If a guideline change moved the value to 37.8, two places would have had to be changed — with a high probability of overlooking one of them |
| **Root Cause** | A contradiction of our own maintainability claim from 08 |
| **Fix** | All thresholds as named constants in `meta.thresholds` ; rules reference `fever` instead of `temp` |
| **Retest** | T2, T19 |

### B-05 • `anticoag` was collected but never used

| | |
|---|---|
| **Severity** | 🟢 low (design/data protection) |
| **Originating layer** | **L2** |
| **Description** | The anticoagulant status fed into no rule at all — even though it is clinically relevant for assessing bleeding |
| **Root Cause** | Collection without a purpose of use. In addition a data protection problem: personal health data collected without a purpose contradict data minimisation |
| **Decision** | Build it in rather than remove it — the clinical value justifies collecting it |
| **Fix** | New rule **R10**: `bleeding >= 1 AND anticoag = 1` → YELLOW |
| **Retest** | T16 |

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### B-06 · Southampton grade IV mapped to RED in the L1 table but to ORANGE in the rules

| | |
|---|---|
| **Severity** | 🟡 medium (inconsistency) |
| **Originating layer** | **L2** — inconsistency between the L1 mapping table and the decision table |
| **Found by** | Test run T15 |
| **Description** | The Southampton mapping in 03 assigned grade **IV (pus)** to RED. The implemented rules, however, escalate to RED only for `wound_state = 4` (the "open / pus" image) or dehiscence (R3); isolated purulent discharge ( `secretion = 3` ) triggers R4 → ORANGE. Two documents made different statements about the same finding |
| **Root Cause** | The mapping table was derived from the Southampton severity scale ("grades IV and V are major complications"), the rules from the clinical urgency of the action required. Both are defensible — but they were never reconciled |
| **Decision** | The **rules** are authoritative; the mapping table has been corrected. Purulent discharge is the primary criterion of a superficial incisional SSI and requires contact **today** (ORANGE) — not the emergency service tonight. RED remains reserved for systemic signs (R1), lymphangitis (R2) and dehiscence or the "open / pus" wound image (R3) |
| **Fix** | Mapping table in 03 §3.4 corrected; rationale documented as challenge C-09 |
| **Retest** | T10, T15 |

### B-07 · Trend rule R12 could never fire — circular dependency

| | |
|---|---|
| **Severity** | 🔴 high (rule without effect) |
| **Originating layer** | **L3/L4** — evaluation order |
| **Found by** | Test run T17 |
| **Description** | The derived element `trend_worsening` compares today's classification with yesterday's. But today's classification is only known **after** the rules have been evaluated — while the derived elements are computed **before**. `classification` was therefore always `null` , `trend_worsening` always `0` , and rule R12 could never fire |
| **Root Cause** | A classic circular dependency: a rule needs its own result as an input. Only visible once the tests were actually executed — a review of the decision table alone would never have found it |
| **Fix** | **Two-pass evaluation** in the L4: the rules run once, the provisional classification is fed back into the scope, the rules run a second time. The worse of the two results wins, so a second pass can escalate but never de-escalate — worst-first also applies across passes |
| **Retest** | T17 |

### B-08 · R12 was unreachable as originally specified

| | |
|---|---|
| **Severity** | 🟡 medium (dead rule) |
| **Originating layer** | **L2** — decision table |
| **Found by** | Test run T17, after B-07 had been fixed |
| **Description** | Even with a working two-pass evaluation, R12 ("deterioration compared with yesterday, yesterday YELLOW") remained unreachable: any deterioration beyond YELLOW already triggers a higher-priority orange or red rule. The condition could only be satisfied in a state that never occurs |
| **Root Cause** | The rule was formulated from the clinical intent ("things are getting worse") without checking against the worst-first cascade whether that state is reachable at all |
| **Fix** | R12 rewritten as a **persistence rule**: `classification = 'yellow' AND last_classification = 'yellow'` → ORANGE. A single abnormal day is tolerable, a persisting abnormal finding is not |
| **Retest** | T17 |
| **Lesson learned** | Every new rule must be checked for reachability against the cascade — added to the L3 review checklist |

## 10.4 Test Protocol

> **Executed on 3 August 2026** against `l3/wundcheck-l3.json` v0.3.3 using the harness `tools/run-tests.mjs`, which uses the same expression engine as the L4 app. Reproduce with `node tools/run-tests.mjs`.

| ID | Checks | Hazard | Expected | Observed | Status |
|---|---|---|---|---|---|
| T1 | R1 fever | H-01 | RED (R1) | `RED (R1) · SG 0` | ✅ passed |
| T2 | R1 boundary | H-01 | RED (>= inclusive) | `RED (R1) · SG 0` | ✅ passed |
| T3 | R3 safety rule | H-01 | RED (image alone escalates) | `RED (R3) · SG V` | ✅ passed |
| T4 | R5 before R6 | H-02 | ORANGE (R5 wins) | `ORANGE (R5) · SG II` | ✅ passed |
| T5 | mild signs, day 2 | H-04 | GREEN | `GREEN (R8) · SG I` | ✅ passed |
| T6 | conditional logic | H-03 | 3 follow-ups hidden | `hidden: true` | ✅ passed |
| T7 | constraint temp | H-05 | input rejected | `rejected: true` | ✅ passed |
| T8 | info rules non-exclusive | — | GREEN + I1 + I2 + I3 | `GREEN (R8) + I1, I2, I3 · SG 0` | ✅ passed |
| T9 | scab reassures | H-04 | GREEN + I4 | `GREEN (R8) + I4 · SG 0` | ✅ passed |
| T10 | worst-first vs reassurance | H-02 | ORANGE (R4) + I4 — warning beats reassurance | `ORANGE (R4) + I4 · SG IV` | ✅ passed |
| T11 | B-02 fixed | H-01 | ORANGE (R9) | `ORANGE (R9) · SG 0` | ✅ passed |

| ID | Checks | Hazard | Expected | Observed | Status |
|---|---|---|---|---|---|
| T12 | B-03 fixed | H-01 | RED (R2) | `RED (R2) · SG 0` | ✅ passed |
| T13 | B-01 fixed | H-01 | YELLOW (R6b), not green | `YELLOW (R6b) · SG II` | ✅ passed |
| T14 | B-01 counter-check | H-04 | GREEN (R6a does not fire) | `GREEN (R8) · SG I` | ✅ passed |
| T15 | Southampton derivation | H-02 | grade IV -> ORANGE (R4) — see B-06 | `ORANGE (R4) · SG IV` | ✅ passed |
| T16 | B-05 fixed | — | YELLOW (R10) | `YELLOW (R10) · SG 0` | ✅ passed |
| T17 | persistence rule R12 | H-01 | ORANGE (R12) after a yellow yesterday | `ORANGE (R12)` | ✅ passed |
| T18 | discharge duration R11 | H-01 | ORANGE (R11) | `ORANGE (R11) · SG III` | ✅ passed |
| T19 | threshold maintainability | — | ORANGE after threshold change | `GREEN -> ORANGE (R5)` | ✅ passed |

**19 passed, 0 failed, 19 total**

Boundary values are covered from both sides: `temp` 38.9/39.0/39.1 · `days_post_op` 2/3/4 · `pain` 6/7 · `discharge_duration_days` 2/3.

**The first run failed 3 of 19 test cases.** All three were genuine defects, not harness errors — they became bugs B-06, B-07 and B-08. Two of them (B-07, B-08) were invisible to a pure document review and only surfaced once the rules were actually executed — which is precisely the point of the exercise.

**Boundary value tests** cover both sides: `temp` 38.9/39.0/39.1 · `days_post_op` 2/3/4 · `pain` 6/7 · `discharge_duration_days` 2/3.

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## 10.5 FHIR Conformance Checks

The decision logic is not the only thing that has to be correct — so does the interface. `tools/check-fhir.mjs` extracts the bundle builder straight out of the L4 and checks the generated transaction bundle against the R4 shape rules we depend on. Reproduce with `node tools/check-fhir.mjs`.

| # | Check | Result |
|---|---|---|
| F01 | Bundle.resourceType = Bundle | ✅ |
| F02 | Bundle.type = transaction — `transaction` | ✅ |
| F03 | every entry has request.method + url | ✅ |
| F04 | QuestionnaireResponse present | ✅ |
| F05 | QR.status = completed | ✅ |
| F06 | QR.questionnaire is versioned canonical — `http://wundcheck.example/Questionnaire/wundcheck | 0.3.2` |
| F07 | QR.subject set — `Patient/example` | ✅ |
| F08 | QR items all have exactly one answer | ✅ |

| # | Check | Result |
|---|---|---|
| F09 | QR answers are typed (not all valueString) — `valueString, valueDate, valueCoding, valueInteger, valueDecimal` | ✅ |
| F10 | QR item count matches answered elements — `33 / 33` | ✅ |
| F11 | Observation present | ✅ |
| F12 | Observation.status = final | ✅ |
| F13 | Observation.code has a SNOMED coding | ✅ |
| F14 | Observation carries the L3 version — `0.3.2` | ✅ |
| F15 | Observation carries the triggering rule | ✅ |
| F16 | Communication generated for ORANGE | ✅ |
| F17 | Communication.priority = urgent for ORANGE — `urgent` | ✅ |
| F18 | Communication has a recipient — `Organization/practice` | ✅ |
| F19 | endpoint comes from the L3, not the code | ✅ |
| F20 | no Communication when GREEN | ✅ |

**20 passed, 0 failed, 20 total**

Check F19 is the interesting one: it asserts that the endpoint URL appears **only** in the L3, not in the app source. Should anyone ever hardcode an endpoint into the L4, this test fails — the maintainability claim is thereby machine-enforced rather than merely asserted in prose.

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## 10.6 No-Hardcoding Check

The central claim of this project — *the L4 contains no clinical content* — is easy to assert and easy to break. We therefore made it a test. `tools/check-hardcoding.mjs` scans the application source (with the embedded L3 payload stripped out) for every element id, section id, rule id, threshold name and value, code system URI, the FHIR endpoint, and a list of clinical vocabulary. Any hit fails the check.

| Check | Result |
|---|---|
| Application source scanned | `479` lines |
| Terms tested (element ids, section ids, rule ids, thresholds, code systems, endpoint, clinical vocabulary) | 72 |
| Clinical content found in the L4 | ✅ none |

**The check found two real leaks when first run**, both of which have been fixed:

| Leak | Fix |
|---|---|
| `Observation.code` carried SNOMED 225552003 "Assessment of wound" directly in the app, and the result badge read `derived.southampton_grade` by name | Both declared in the L3 ( `meta.fhir.observation_code` , `meta.display.result_badge` ); the app now iterates over the declaration |
| The "New day" function reset the sections `general` and `wound` by their ids — the app knew which sections were clinical | Reset now derives from the sections' `once` flag; the app no longer knows any section by name |
| Terminology system URIs ( `http://snomed.info/sct` ...) were a lookup table in the app | Moved to `meta.terminology_systems` |

This is the same pattern as B-06: a claim made in prose that the implementation did not actually keep. The difference is that this one can no longer drift silently — the check runs with the test suite.

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## 10.7 Step 4 — Residual Risk

> Day 5 explicitly names as a characteristic of sufficient validation: *"Explicit acknowledgment of residual risk and monitoring strategy."*

**What we have not tested:**

**(a) Behaviour with real patients.** All tests are desk-based. We have not had a single real wound assessed.

**(b) The reliability of image assignment by medical laypersons.** This is the **greatest untested risk of our entire approach**. The translation of a clinical grading into an image selection (C-01) is the central design idea — and it is unvalidated. In particular: erythema appears visually very different on different skin types; our graphics depict only one skin type.

**(c) Offline and synchronisation behaviour across several days.** The trend rule R12 and the derived elements `last_classification` / `trend_worsening` presuppose an uninterrupted data history. What happens when there are gaps has not been tested.

**(d) Transferability to all body regions.** The scope covers all elective wounds with primary closure — yet our graphics show only one generic wound type. An abdominal wound looks different from a wound on a finger.

**(e) Clinical sign-off.** All eight L1→L2 decisions were taken by a group of students, not by clinical professionals.

### How We Address This

| Residual risk | Measure |
|---|---|
| (a), (b) | Usability study during the pilot phase; parallel assessment by the patient and a healthcare professional to determine agreement |
| (b), (d) | **Most important safety indicator in post-deployment monitoring:** follow-up of all cases that presented to a physician within 72 hours despite a GREEN classification |
| (c) | Explicit tests of the trend logic with data gaps before the pilot |
| (d) | Extension of the graphics with body-region-specific variants; until then a scope notice in the app |
| (e) | Clinical review of all `meta.thresholds` and decision rules as a precondition for the pilot |

Details: 11 Monitoring & Evaluation

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## 10.8 "Good enough validation" — Self-Assessment

Day 5 names six characteristics. Where we stand:

| Characteristic | Status |
|---|---|
| Documented risk analysis identifying highest-harm scenarios | ✅ Hazard table H-01...H-06 |

| Characteristic | Status |
|---|---|
| Evidence of testing on critical clinical paths | ✅ 19 of 19 test cases executed and passed; 8 defects found and fixed |
| Clinical expert sign-off on rule logic and edge cases | ❌ **missing — documented as a limitation** |
| Usability evidence that users can act correctly | ❌ **missing — greatest residual risk** |
| Post-deployment monitoring plan with incident response | ✅ 11 M&E |
| Explicit acknowledgment of residual risk | ✅ Section 10.7 |

**Four out of six.** The two missing points are exactly those that separate a student prototype from a deployable system — and both require access to clinical staff and to real patients.

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**Related documents:** 06 Decision Logic · 07 Challenges · 11 M&E